

Data Readiness Gate Report

Model Name: HC Manuscript Evaluator

Model ID: HC-QG-001

Tier: Tier 1 (High Risk)

Date: April 2026

1. Objective

This report assesses whether the dataset prepared for validation of the HC Manuscript Evaluator meets the required quality, completeness, and fairness standards defined in the project governance framework.

The purpose of this gate is to ensure that model validation is conducted on data that is reliable, representative, and free from material bias that could invalidate downstream results.

2. Data Quality Assessment

2.1 Missingness

- **Observed:** 0.00%
- **Threshold:** 0%
- **Status:** PASS

Assessment:

The dataset contains no missing values across required fields. This meets the strict completeness requirement for Tier 1 systems and supports reliable downstream evaluation.

2.2 Bias Assessment – Demographic Parity Difference (DPD)

- **Female Average Score:** 5.00
- **Male Average Score:** 5.75
- **DPD:** 0.75
- **Threshold:** ≤ 0.10
- **Status:** FAIL

Assessment:

The observed disparity of 0.75 significantly exceeds the acceptable threshold, indicating a material imbalance in model scoring behavior across demographic groups. This level of

deviation suggests a high risk of systematic bias and renders the dataset unsuitable for fair model evaluation in its current state. Triggers immediate bias investigation when score gaps exceed 0.10, preventing discriminatory publishing decisions.

2.3 Bias Assessment – Disparate Impact Ratio (DIR)

- **DIR:** 0.870
- **Threshold:** 0.80 – 1.25
- **Status:** PASS

Assessment:

The dataset meets the DIR requirement, indicating that outcomes fall within acceptable proportional bounds. However, this does not offset the failure observed in DPD, as both metrics must be satisfied for fairness validation. Proves legal compliance with anti-discrimination law (four-fifth rule) to defend against lawsuits.

2.4 Score Variability

- **Mean:** 5.38
- **Standard Deviation:** 0.63
- **Coefficient of Variation (CV):** 0.117

Assessment:

Score variability is within a reasonable range, indicating consistent scoring behavior without excessive dispersion or collapse. This supports statistical stability but does not mitigate fairness concerns. Confirms the AI is stable enough for production use, not randomly broken or giving wildly inconsistent scores.

3. Overall Assessment

Dimension	Status	Commentary
Completeness	PASS	No missing data
Fairness (DPD)	FAIL	Significant demographic disparity detected
Fairness (DIR)	PASS	Within acceptable bounds
Statistical Stability	PASS	Acceptable variability

4. Decision

Decision: NO-GO

The dataset fails the Data Readiness Gate due to a material breach of the Demographic Parity Difference (DPD) threshold. As a Tier 1 system, fairness requirements are non-negotiable, and this level of bias invalidates the dataset for use in model validation.

Proceeding with validation under these conditions would introduce a high risk of producing misleading or non-compliant model performance results.

5. Required Remediation Actions

- Conduct root cause analysis to identify the source of demographic score disparity
- Review and rebalance dataset composition across demographic groups
- Reassess labeling guidelines and scoring consistency
- Regenerate and re-evaluate the dataset against fairness thresholds prior to resubmission

6. Approval Status

Role	Decision	Signature	Date
Model Governance Lead	NO-GO	_____	April 2026
Model Risk Management (MRM) Lead	NO-GO	_____	April 2026
Editorial Director (Business Owner)	NO-GO	_____	April 2026